CHAPTER 1

Introduction

While it may be tempting to frame the concept of protein function in teleological terms, in reality it is pragmatically defined by the distinct and diverse experimental tools available to a biologist. For example, experimentalists can detect how a protein affects the rate at which a reaction catalyzes through enzymatic assays. Yeast 2-hybrid assays can indicate binding between pairs of proteins. Microarrays can be used to approximate the expression levels of a gene in a tissue sample. Results from different experiments are then brought together in order to paint a picture of how a protein behaves, usually culminating in bestowing upon the protein a particular function, most commonly, in the form of a Gene Ontology (GO) term annotation It is both the widespread application of GO terms, and its standardization of terminology, that has made much of the quantitative analysis of protein function possible.

I would consider restructuring each chapter as

1. Intro
2. Methods
3. Results
4. Discussion

all references need to be fixed!

CHAPTER 2

Analysis of protein function and its prediction from sequence alignments

1. Introduction

Protein function is a multi-scale concept that reflects "everything that happens to or through a protein",[Rost et al., 2003] and is typically considered from the biochemical, biological, and phenotypic perspectives. [Bartlett et al., 2003] From the biochemical, or molecular, perspective a protein may be a kinase, whereas in terms of its biological function this kinase can be involved in numerous processes, such as cell cycle regulation or cell-cell signaling. Two proteins with the same molecular function may be involved in drastically different biological processes, and conversely, the set of proteins associated with a particular biological process will generally be drawn from a wide range of molecular functions. From the phenotypic viewpoint, a protein is generally associated with variation in observable physical or behavioral traits. For example, kinase variants or mutants may be responsible for disease. Adding another level of complexity to the study of function is the fact that a protein's molecular function, while generally considered to be a static notion, is modulated by a particular cellular context, e.g. the presence of other molecules, or properties of the physical environment, e.g. temperature.[Mohan et al., 2009]

Several classification systems have been proposed to standardize functional annotation and to facilitate computation. With few exceptions, these classification systems usually take on the structure of hierarchical ontologies. Enzyme Commission (EC) numbers [NCotIUoBaMB, 1992] and the MIPS functional catalogue [Ruepp et al., 2004] are two well-accepted schemes; however, the most commonly used functional classification is the Gene Ontology (GO). GO provides three hierarchical classifications as directed acyclic graphs: Molecular Function Ontology (MFO), Biological Process Ontology (BPO), and Cellular Component Ontology.[Ashburner et al., 2000] With respect to defining a particular gene's phenotype, the existing classifications are species-specific (e.g. the Human

Phenotype Ontology [Robinson and Mundlos, 2010]) and predominantly constructed to address human disease. The Unified Medical Language System, [Bodenreider, 2004] for example, incorporates a number of vocabularies together with semantic relationships between terms, mainly for the purpose of defining associations between genes and medical disorders.

Experimental studies of protein function have been steadily accumulating over the years. Currently, there are about 50,000 proteins with at least one experimentally annotated GO term from MFO or BPO in the GO database and Swiss-Prot [Bairoch et al., 2005a] combined. However, owing to the numerous sequencing projects, [Liolios et al., 2008] the gap between annotated and non-annotated proteins has exceeded two orders of magnitude, and will only get wider. In addition, annotated proteins may not be sufficiently covered, both with respect to the resolution of functional annotations and the inclusion of other yet unknown functions. Thus, it is important to develop algorithms capable of accurately predicting protein function.

Historically, sequence-based inference was the first strategy used to predict protein function, even if most studies at the time avoided explicitly relating homology and function.[Doolittle, 1986] Global and local sequence alignments were used to query sequence databases for similarities with a target protein. With the accumulation of experimentally determined protein functions, the most similar annotated sequences have traditionally been used to infer function.[Rost et al., 2003] More advanced methods exploited predicted physicochemical properties,[Jensen et al., 2002, 2003] evolutionary relationships,[Engelhardt et al., 2005, Pellegrini et al., 1999, Bandyopadhyay et al., 2006]or the structure of functional ontologies in order to achieve different confidence levels at different ontological terms.[Barutcuoglu et al., 2006, Hawkins et al., 2006, Martin et al., 2004] Microarrays, protein-protein interaction networks, protein structures, and predicted ligands have also been exploited.[Deng et al., 2003, Hermann et al., 2007, Lee et al., 2004, Troyanskaya et al., 2003, Letovsky and Kasif, 2003, Nabieva et al., 2005, Pal and Eisenberg, 2005, Pazos and Sternberg, 2004, Song et al., 2007, Vazquez et al., 2003] However, most of these methods are limited to a few organisms where such data are available. One way or another, sequence alignment-based inference is the cornerstone of functional inference and is the focal point of this study.

Sequence alignment-based transfer of function has been thoroughly studied in the last decade, predominantly for enzymes.[Addou et al., 2009, Devos and Valencia, 2000, Rost et al., 2003, Tian and Skolnick, 2003, Todd et al., 2001, Wilson et al., 2000] The results of these studies indicate that at

least 60% sequence identity, and more likely closer to 80%, is required for the accurate transfer of the third level of EC classification. More sophisticated approaches were proposed as well: the GOtcha method was developed in order to take sequence alignment scores between a query protein and a functionally annotated database and overlay them on the functional ontology, cumulatively propagating such scores toward the root ~~of the ontology.~~[Martin et al., 2004] PFP refined this technique by incorporating PSI-BLAST alignments at very low significance levels and conditional probabilities that a protein is associated with pairs of functional terms.[Hawkins et al., 2006] Other methods such as ProtFun,[Jensen et al., 2003] ConFunc,[Wass and Sternberg, 2008] GOsling,[Jones et al., 2008] and GOstruct [Sokolov and Ben-Hur, 2010] were developed for high-throughput prediction tasks. Finally, phylogenetic methods attempt to exploit particular evolutionary relationships within a gene family. [Brown and Sjolander, 2006, Eisen, 1998] Methods such as SIFTER [Engelhardt et al., 2005] or ortholog identification methods [Remm et al., 2001] belong in this category. Several recent reviews provide good perspectives on protein function prediction at all scales.[Dalkilic et al., 2008, Friedberg, 2006, Kann, 2007, Laskowski and Thornton, 2008, Lee et al., 2007, Punta and Ofran, 2008, Rentzsch and Orengo, 2009, Rost et al., 2003]

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This study addresses protein function at the molecular function and biological process levels, as defined by the GO consortium, and functional inference from protein sequence alone. We analyze the power of functional transfer of GO terms using sequence alignments and develop a method for probabilistic inference of GO terms using supervised learning. Our algorithm, Functional ANNotator (FANN), employs multi-output artificial neural networks. We show that in the GO annotation task, FANN-GO outperforms standard sequence alignment methods and GOtcha.

2. Results

2.1. Multi-functional proteins. It is well-known that some proteins have the ability to carry out more than one function, or "moonlight".[Jeffery, 1999, 2009] Such abilities can be facilitated by either having multiple functional domains, a single domain which binds multiple partners, or by different behavior upon post-translational modifications or change in physiological conditions.[Jeffery, 2004, 2009] Here we systematically analyze multi-functionality with respect to Molecular Function (MFO) and Biological Process Ontologies (BPO) in GO. As an approximation of distinct functions

we only consider the number of experimentally determined leaf GO terms associated with each protein. A GO term g was included in the count if no other term associated with the protein had g as its more general term in the ontology. For example, if a protein is associated with the term "protein binding", the term "binding" is not counted as distinct function because it is a generalization of "protein binding"; however, the term "nucleic acid binding" would be counted because neither of the two terms is a generalization of the other.

Figure 2.1 shows the distribution of the number of leaf terms associated with each functionally annotated sequence for both ontologies. In total, 26,707 sequences were included in the MFO analysis and 29,118 sequences were included in the BPO analysis (see Section 4.2). Only experimental evidence, traceable author's statement, or curator's inference were considered (exclusion of traceable author's statement and curator's inference resulted in very similar distributions; data not shown). The plots show greater diversity in a protein's participation in a biological process than its ability to carry out distinct molecular functions. About 66% of the proteins experimentally annotated by MFO terms have only 1 leaf term, with no protein having more than 14. On the other hand, only 44% of proteins have only 1 BPO leaf term, with 6 proteins having 50 terms or more. This is consistent with the expectation that biological processes are governed more by the context in which a protein is utilized, and less by the physicochemical abilities of the protein. Interestingly, both the molecular function and biological process ontologies show a scale free-like decrease in the probability that a protein is associated with an increasing number of functional terms. In the context of protein function prediction, these distributions suggest there is a level of sophistication that should be required from a computational method with respect to its outputs.

We also analyzed the relationship between the number of molecular functions and biological processes a protein is associated with. It seems intuitive to postulate that proteins with the ability to carry out multiple molecular functions should be more easily utilized in multiple different contexts, giving rise to an association with more biological processes. Out of 19,240 proteins in the intersection of data sets for molecular function and biological process, we found a Pearson correlation coefficient of 0.261 between the numbers of associated leaf terms in the two ontologies. While this correlation may seem weak, we determined that this value is statistically significant by using a permutation test where the numbers of MFO and BPO terms were permuted in the data set of proteins. We carried out 100,000 such permutations and did not find any cases in which the correlation coefficient was

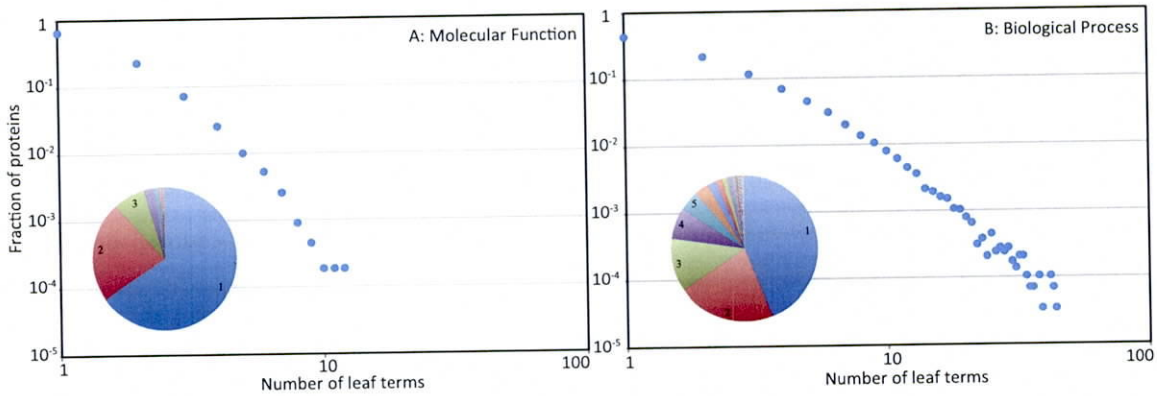

Figure 2.1. The distribution of the number of leaf terms in (A) Molecular Function and (B) Biological Process ontologies. The x-axis represents the number of leaf terms associated with a protein; the y-axis represents the fraction of proteins in the data set with the given number of leaf terms. Both axes are in log10 scale. The inset in each panel provides a pie chart that corresponds to each plot.

0.261 or greater (the mean correlation coefficient for permuted data was 6.1×10^{-6} and standard deviation was 7.2×10^{-3}).

While, in general, a protein performing multiple molecular functions is associated with multiple biological processes, we conducted further analysis of proteins associated with a single term from one ontology and multiple terms from the other. For example, among the proteins that have only one MFO leaf term, but multiple BPO terms, we found that receptor binding terms such as "chemokine receptor binding" and "cytokine receptor binding" are the most enriched ($P < 1.0 \times 10^{-7}$; Binomial test). On the other hand, there are also cases in which proteins that are associated with only one leaf term in BPO are related to multiple MFO terms. Such BPO terms are almost all related to metabolic processes, e.g. "cellular metabolic process", "primary metabolic process", etc. ($P < 1.0 \times 10^{-7}$; Binomial test). When analyzing this class of proteins we also found that some terms did not occur as often as expected. When looking at the class of proteins with one MFO term and multiple BPO terms we found that "catalytic activity" was depleted. While sequences are annotated with this term 28% of the time in the whole data set (29% in the set of all proteins with single MFO leaf term), it only occurs 17% of the time when we only consider sequences with exactly one MFO term but 3 or more BPO terms associated with them. Similarly, proteins having 1 BPO leaf term, but

few MFO terms are usually involved in reproduction. We note that this data needs to be interpreted with caution, because MFO and BPO terms are incomplete for most proteins and also because there may exist biases in ways current functions are acquired.

Finally, we looked at the relationship between the number of Pfam domains [Finn et al., 2008] found in a protein and the number of BPO and MFO terms it was associated with. We found that the Pearson correlation between the number of MFO leaf terms and Pfam domains (detected using a threshold of 0.001) was much larger than that of BPO leaf terms and Pfam domains (0.126 and 0.090 respectively). While, again, these correlation values might seem small, we found both were significant using the permutation test.

2.2. Transfer of function by sequence similarity. We evaluated the performance of simple function transfer between similar sequences. The following steps were taken: for each range of pairwise global sequence identities, a target protein received all functional terms from each experimentally annotated protein within the given sequence identity range (regardless of the species). For each target sequence with more than one match in the given identity range the precision and recall (see Section 4.5) were calculated as averages over all pairs. Then, the precision and recall for the entire data set are reported (Figure 2.2) as averages of the averaged precisions and recalls calculated for each sequence covered (a leave-one-out estimation). We also report the coverage for each range of sequence identity as the fraction of proteins with at least one other annotated sequence whose pairwise sequence identity falls within the defined range. Finally, for all covered sequences we report the percentage of perfect annotations, i.e. the percentage of all pairwise annotations in a given bin where both precision and recall were 1.

The results shown in Figure 2.2 suggest that using global sequence identity for the transfer of functional annotations is only moderately accurate (pairwise local alignments performed similarly; data not shown). Surprisingly, even for 100% identity transfer of function does not achieve either precision or recall of 1 (only 17% of identical sequence pairs had perfect transfer of MFO and 12% of BPO terms). This relatively low rate of perfect annotations among perfect matches, and similarly in the remaining identity bins, in our data set is caused by three factors: (i) sparsity of database annotations, where proteins are incompletely annotated with respect to functionality and also specificity of annotation; (ii) database errors, caused by incorrect interpretation of experiments or by curation errors, [Brenner, 1999, Schnoes et al., 2009] and (iii) organismal context, where the

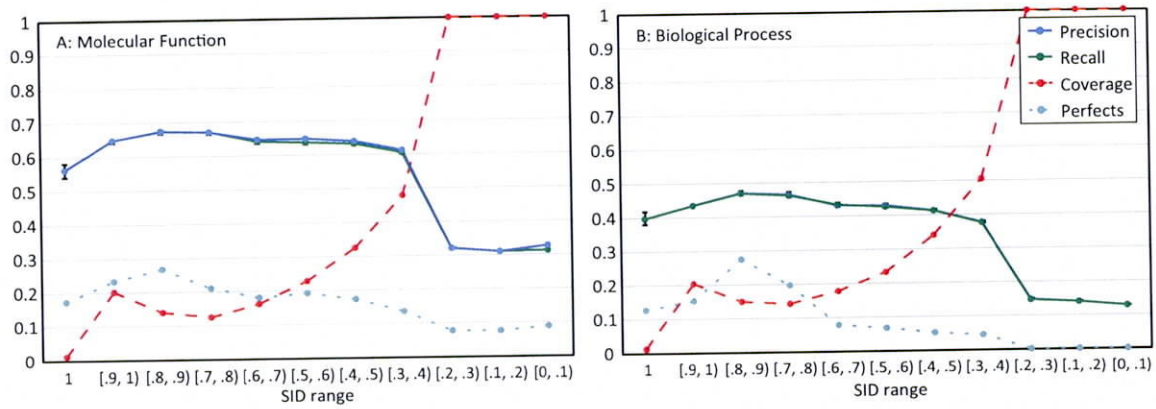

Figure 2.2. Accuracy of function transfer using global pairwise sequence identities (A: Molecular Function; B: Biological Process). For each sequence identity range (x-axis), shown are the average precision (blue solid line) and recall (green solid line) of function transfer by pairwise similarity. The teal dotted line represents the percentage of pairs with perfect annotations (e.g. both precision and recall equal to 1). The red dashed curve represents the percent of proteins that have pairwise matches (annotated with GO terms; experimental evidence code) in a given range. The error bars represent 95% confidence intervals.

difference between two organisms influences a particular functional role of individual proteins, even at 100% sequence identity.

In general, transferring MFO annotations is more accurate than transferring BPO annotations, which has previously also been observed by Rogers and Ben-Hur in a different prediction scenario.[Rogers and Ben-Hur, 2009] This is probably a result of the fact that MFO terms are less dependent upon cellular, tissue, or organismal context, but also that the topological properties, including the average branching factor, the number of terms, and the average depth of a leaf node, between the two ontologies differ (data not shown). For both ontologies the precision of transferring predictions rapidly decreases once the $20 - 30\%$ identity is reached. This range of sequence identity has been termed the "twilight zone" for the inference of protein structure from sequence.[Rost, 1999] In the context of function transfer such a twilight zone cannot be clearly defined (or rather should be extended to the entire range $30 - 100\%$), with the range below 30% being one where function transfer breaks down completely ("midnight zone"). We also evaluated an alternative approach

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where functional terms from all matches in a given identity range were transferred to the target protein and found a very similar level of precision but an increased recall (Figure 2.7).

rename to 2.3

The quality of function transfer is highly dependent on the particular class of protein. We show this by splitting the MFO/BPO data sets into enzymes (proteins annotated with term "catalytic activity"), and non-enzymes. As seen in Figure 2.3, the ability to transfer molecular functions to these two classes of proteins was noticeably different and most likely points to a higher quality of functional annotations for enzymes.

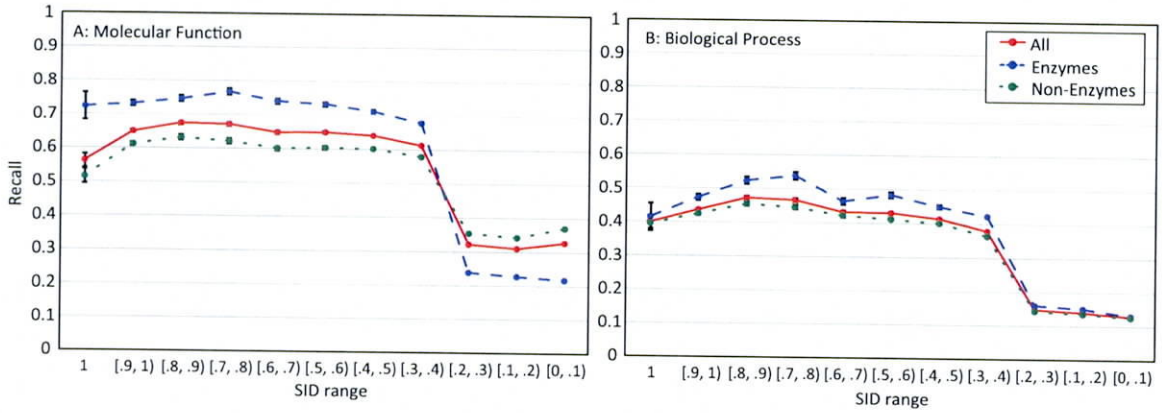

Figure 2.3. Accuracy of function transfer using global pairwise sequence identities for enzymes vs. non-enzymes (A: Molecular Function; B: Biological Process). For each sequence identity range (x-axis), shown is the average precision of function transfer by pairwise similarity (blue dashed line: enzymes, green dotted line: non-enzymes; red solid line: combined). The error bars represent 95% confidence intervals

Another interesting trend in the data is the fact that the quality of annotations transferred between sequences from the same species is higher than that obtained between different species. This can be seen by comparing the precision/recall curves in Figure 2.4 obtained by only considering pairs of sequences from the same species ("within" curve, blue line), and only considering pairs of sequences from different organisms ("between" curve, green line). A similar trend has been previously observed on protein-protein interaction data by [Mika and Rost, 2006].

2.3. Quality of non-experimental annotations in Swiss-Prot. Protein databases such as GO or Swiss-Prot contain a number of functional annotations supported by non-experimental evidence codes. We aimed to assess the quality of such annotations by analyzing non-experimental

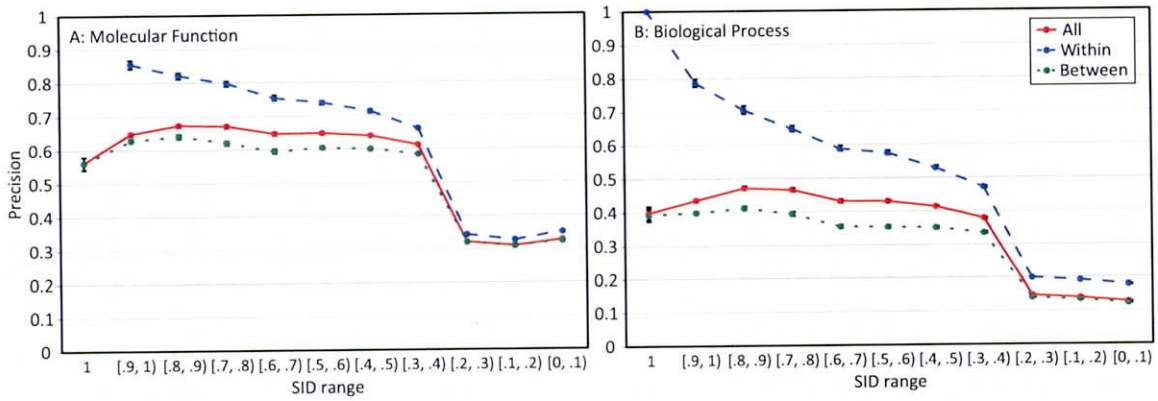

Figure 2.4. Accuracy of function transfer using global pairwise sequence identities from the same or different organism (A: Molecular Function; B: Biological Process). For each sequence identity range (x-axis), shown is the average precision of function transfer by pairwise similarity (blue dashed line: same organism, green dotted line: different organism; red solid line: combined). The error bars represent 95% confidence intervals.

annotations for the proteins in Swiss-Prot (v10.0-v15.0) that in a later release (v15.15) accumulated experimental annotations. Figure 2.5 shows the quality of annotations by non-traceable author statement (NAS) or inferred from electronic annotation (IEA) evidence codes (the remaining non-experimental codes did not contain enough sequences). Using the current annotations (v15.15) as true function, the precision and recall of each protein's annotation were calculated (Figure 2.5).

As shown in Figure 2.5, the quality of electronic annotations is consistently better than that of non-traceable author statements for MFO, while the trend is reversed for BPO. Interestingly, the precision of the electronic annotations in the Swiss-Prot database for MFO exceeds the level achieved by a simple transfer of annotation (Figure 2.2), while the NAS evidence suggest lower confidence levels compared to that of sequence transfer. A historical analysis of BPO annotations suggests that neither electronic annotation nor non-traceable author statements have been at the level of simple transfer of annotation, however the latest major release of Swiss-Prot (v15.0) provides more accurate electronic inference than transfer by sequence similarity.

2.4. Performance of Functional ANNotator (FANN). In this work we present a simple classification method, FANN-GO, for predicting GO terms from MFO and BPO. The model is based

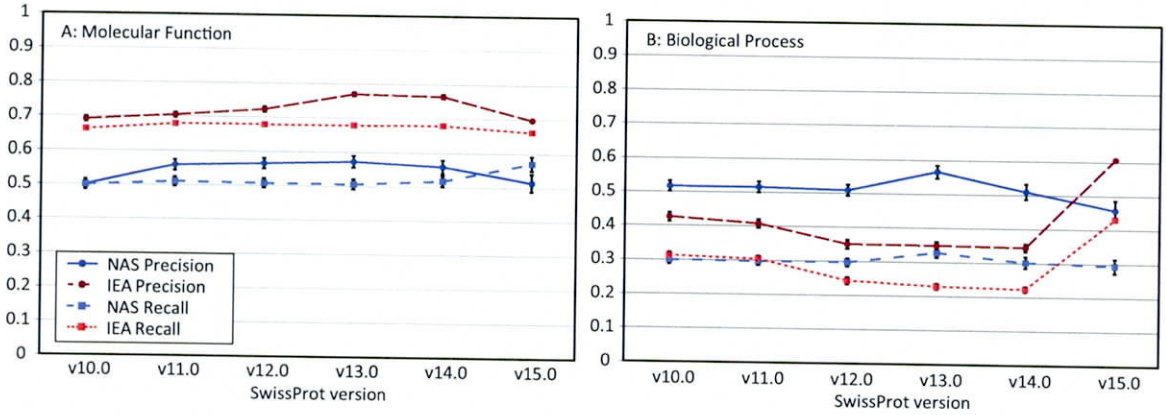

Figure 2.5. Accuracy of function transfer for the Swiss-Prot database (A: Molecular Function; B: Biological Process). X-axis represents a different version of Swiss-Prot. The dark dashed red line represents the precision of IEA annotations and the dark blue solid line represents the precision of NAS annotations. The light red dotted line represents the recall of IEA annotations, and the light blue dashed line represents the recall of NAS annotations. The error bars represent 95% confidence intervals.

on aligning a target sequence to a database of experimentally annotated proteins and calculating the i-score [Martin et al., 2004] that the protein is associated with each functional term. These scores were then used as inputs to an ensemble of multi-output neural networks that were trained to predict the probability that the protein is associated with each function. A variant of FANN-GO, referred to as FANN-GOspecies, is also presented. This model is based on two groups of inputs, one containing inputs where a target protein is aligned only to proteins from the same species and another where the protein is aligned to proteins from different species.

We compared FANN-GO models with GOtcha as well as transfer by global and local sequence identity. We also implemented a nave method that uses prior probabilities of functional terms as prediction scores for all target proteins. That is, the score for each function, for all proteins, is simply the relative frequency of that term occurring in the (training) data set. Performance of all classification models was assessed using 10-fold cross-validation on all functional terms associated with 50 proteins or more (mainly for the purpose of stable accuracy estimation). In doing so, we reduced the number of terms associated with the sequences in the data set for molecular function to

344 and biological process to 1,788. For each ontology all methods were evaluated using the same data sets.

2.6

Figure 6 shows the performance of our two different supervised methods, FANN-GO (blue line), and FANN-GOspecies (green line) compared to an in-house implementation of the GOtcha method (red line), and two methods utilizing transfer of annotations based on pairwise sequence identity: Global-SID (purple line) based on global sequence alignments, and Local-SID (teal line) based on local BLAST alignments. These methods were benchmarked against a Naïve classifier (orange line).

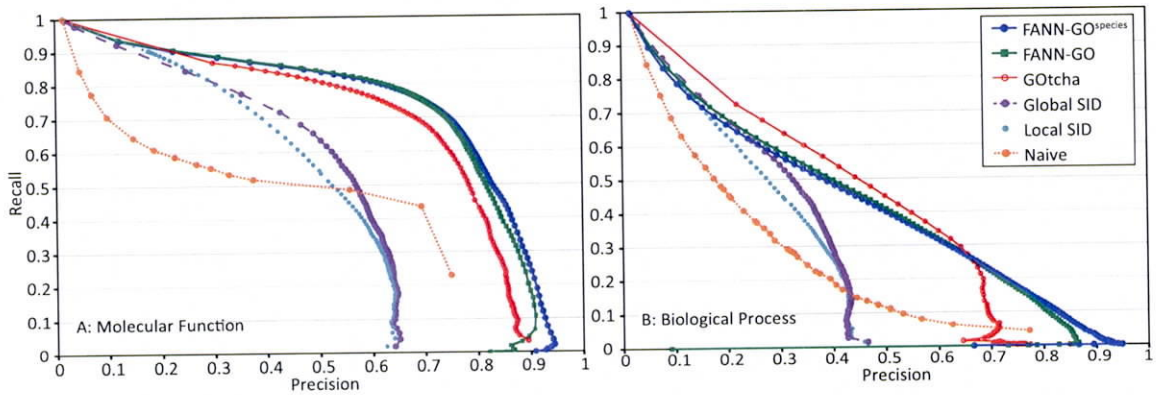

Figure 2.6. Precision-recall curves for several GO-term prediction algorithms (A: Molecular Function; B: Biological Process). The curves were generated by shifting the decision threshold t from 0 to 1 and considering all terms with scores $> t$ as predicted functions. The decision threshold for the sequence alignment curves (Global-SID, Local-SID) was the pairwise sequence identity (SID). The Naïve curve (orange dotted) was generated by assigning all proteins the most common functional term, then the second most common, etc. FANN-GO^{species} is represented by the solid blue line. FANN-GO is represented by the solid green line with square markers. GOtcha is represented by the solid red line with red circular markers. Global-SID is represented by the purple dashed line. Finally, Local-SID is represented by small teal markers only.

italic

As shown in Figure 2.6A, FANN-GO outperformed GOtcha in both precision and recall when annotating sequences with MFO terms. Both methods outperformed simple functional transfer based on sequence identity, and nave predictions. Interestingly, annotating sequences with the most